$$\text{Slip } S = \frac{N_S - N}{N_S}$$

$$\% \text{ Slip} = \frac{N_S - N}{N_S} \times 100$$

8.5.4 Synchronous Motor

A motor which is running in its synchronous speed is called as synchronous motors.

1. For a given frequency, the synchronous motor runs at a constant average speed, whatever the load.
2. Synchronous motor can be operated over a wide range of power factors, both lagging and leading.
3. The synchronous motor is not a self-starting motor.
4. The changes in applied voltage do not affect synchronous motor torque.
5. For synchronous motor, D.C excitation is required.
6. Synchronous motors are usually more costly and complicated

Applications of Synchronous Motor

1. Power factor correction
2. Constant speed, constant load drives
3. Constant voltage

8.6 STEPPER MOTOR

A stepper motor is an electromechanical device which converts electrical pulses into discrete mechanical movement.

8.6.1 Step Angle

Step angle is defined as the angle which the rotor of a stepper motor moves when one pulse is applied to the input of the stator.

The position of a motor is decided by the step angle and is expressed in degrees

$$\text{Step angle, } (\theta_s) = 360/mn$$

m - number of phases.
n – number of rotor teeth.

Stepper motor can be divided into the following three categories.

1. Variable Reluctance (VR) stepper motor
2. Permanent Magnet (PM) stepper motor
3. Hybrid stepper motor

8.6.2 Variable Reluctance Stepper Motor

It is the most basic type of stepper motor. The VR stepper motor has stator and rotor. The stator windings are wound on the stator poles. The rotor carries no windings. Rotor poles are of a ferromagnetic material. The rotor is a salient pole type. This motor may be single stack or multi stack type. This is called variable reluctance motor because the reluctance of the magnetic circuit formed by the rotor and stator teeth varies with the angular position of the rotor. The direction of motor rotation is independent of the polarity of the stator current.

8.6.3 Permanent Magnet Stepper Motor

In this motor also stator winding are wound on the stator poles. The stator is multipolar. The rotor is generally cylindrical and rotor poles are permanently magnetized. The direction of motor rotation depends on the polarity of the stator current.

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